how the values of the approximation improve as Δx is made smaller. The values of the second derivative, whose analytical value is 2, show an oscillatory behavior due to round-off. The first derivative values converge monotonically to zero.

Program 1

```
ZZJOB 5                                  CSC  001    GERALD, C. F.
ZZFORX5
*LIST PRINTER
C   PROGRAM TO COMPUTE DERIVATIVES BY FINITE DIFFERENCE FORMULAS
C   DEFINE THE FUNCTION BY AN ARITHMETIC STATEMENT FUNCTION
      F(X) = X**2*EXPF(X)
C   READ IN X VALUE AND INITIAL VALUE OF DELTA X
      READ 100, X, H
C   PRINT HEADING
      PRINT 200, X
C   COMPUTE DERIVATIVES AND PRINT.  DECREASE H AND REPEAT
      FO = F(X)
      DO 10 I = 1,10
      FR = F(X + H)
      FL = F(X - H)
      DX = (FR - FL)/(2.*H)
      DDX = (FR - 2.*FO + FL)/H**2
      PRINT 201, H, DX, DDX
   10 H = 0.1*H
      CALL EXIT
  100 FORMAT (2F10.0)
  200 FORMAT (52H DERIVATIVES COMPUTED BY CENTRAL DIFFERENCE FORMULAS /
     1 8H AT X = , F5.2 // 4X, 1HH, 13X, 2HDX, 17X, 3HDDX /)
  201 FORMAT (1H ,E8.1, 5X, E14.7, 5X, E14.7)
      END
0.0       0.1
ZZZZ
DERIVATIVES COMPUTED BY CENTRAL DIFFERENCE FORMULAS
AT X =  0.00

   H              DX                DDX
1.0E-01     1.0016675E-02     2.0100083E+00
1.0E-02     1.0000150E-04     2.0000999E+00
1.0E-03     9.9995000E-07     2.0000009E+00
1.0E-04     1.0000000E-08     2.0000000E+00
1.0E-05     9.9500000E-11     1.9999999E+00
1.0E-06     9.5000000E-13     1.9999999E+00
1.0E-07     5.0000000E-15     1.9999999E+00
1.0E-08     0.0000000E-99     2.0000000E+00
1.0E-09     0.0000000E-99     2.0000000E+00
1.0E-10     0.0000000E-99     2.0000000E+00
```

PROBLEMS

Section 1

1. The following table is for sin θ, with θ given in degrees. Find $d(\sin\theta)/d\theta$, at $\theta = 22°$, correct to four decimals. How many terms of Eq. (1.4) are required? Discuss your findings.

θ	sin θ	θ	sin θ
20	0.34202	25	0.42262
21	0.35837	26	0.43837
22	0.37461	27	0.45399
23	0.39073	28	0.46947
24	0.40674		